

PrimePie: Mathematical and Algorithmic Documentation

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Chapter 1

Introduction

PrimePie is a Python number theory library with C++ backend, designed for high-performance computation and symbolic experimentation.

It is structured into three primary components:

1. Analytic Number Theory
2. Algebraic Number Theory
3. Cryptographic Algorithms

The ie project is guided by the following Philosophy:

- **Pythonic interface:** A simple and flexible Python API that feels native and encourages experimentation.
- **High-performance C++ backend:** Core computations are implemented in C++ for speed and scalability.
- **Seamless visualization:** Built-in support for intuitive and customizable visualization of number-theoretic objects and results.
- **Minimal and general abstractions:** Mathematical structures are defined in a general and reusable way, without unnecessary overhead.

1.1 Philosophy

PrimePie is designed with several principles in mind:

1. Data Types – Precision VS. Approximation TODO

2. Parallel Computation – Scalar Logic, Vector Execution All scalar operations in the library are automatically extended to **vectorized operations**, allowing users to apply the same logic to entire collections of values with minimal effort.

For example, consider the method `.power(a, n)` which computes a^n :

- If **a** is a scalar and **n** is a scalar, it returns a^n .
- If **a** is a list $[a_1, a_2, \dots, a_k]$ and **n** is a scalar m , the function returns $[a_1^m, a_2^m, \dots, a_k^m]$.

- If both **a** and **n** are lists of the same length, the function performs element-wise exponentiation:

$$\text{power}([a_1, \dots, a_k], [n_1, \dots, n_k]) \rightarrow [a_1^{n_1}, \dots, a_k^{n_k}]$$

Chapter 2

Algebra

The algebra module is organized around two basic classes of structures: groups and rings. Groups provide a single operation with inverses, while rings combine an additive group structure with a compatible multiplication.

2.1 Groups

A group is a set G equipped with a binary operation

$$* : G \times G \rightarrow G$$

such that the operation is associative, has an identity element e , and every element $a \in G$ has an inverse a^{-1} . In PrimePie this abstract structure is represented by the common operations

`identity,` `inverse,` `operate.`

Powers in a Group

For a group element a and an integer n , the power a^n means repeated use of the group operation:

$$a^n = \begin{cases} e, & n = 0, \\ \underbrace{a * a * \cdots * a}_{n \text{ times}}, & n > 0, \\ (a^{-1})^{-n}, & n < 0. \end{cases}$$

This definition applies to any group, not only to multiplicative groups. For an additive group, a^n means repeated addition of a .

The algorithm used for group powers is binary exponentiation. For positive n , it computes powers in $\mathcal{O}(\log n)$ group operations by repeatedly squaring:

$$a^n = \begin{cases} e, & n = 0, \\ (a^{n/2})^2, & n \text{ even}, \\ a * (a^{(n-1)/2})^2, & n \text{ odd}. \end{cases}$$

Concrete Group Families

The current algebra layer includes modular additive groups, modular multiplicative groups of units, and direct sums of groups.

- The additive group $\mathbb{Z}/n\mathbb{Z}$ uses addition modulo n .
- The multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of residue classes coprime to n , with multiplication modulo n .
- Direct sums combine several groups componentwise:

$$(a_1, \dots, a_k) * (b_1, \dots, b_k) = (a_1 *_1 b_1, \dots, a_k *_k b_k).$$

2.2 Rings

A ring is a set R equipped with addition and multiplication. The additive structure is an abelian group:

$$(R, +, 0),$$

and multiplication is associative and distributes over addition:

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc.$$

The ring structures in PrimePie expose the canonical operations

$$\text{zero}, \quad \text{one}, \quad \text{add}, \quad \text{neg}, \quad \text{mul}.$$

Additive and Multiplicative Powers

Since a ring is an additive group, it has an additive power operation inherited from the group structure. Thus, in a ring,

$$\text{power}(a, n)$$

means repeated addition of a .

Multiplicative exponentiation is a separate operation:

$$\text{mpower}(a, n),$$

which computes

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ times}}$$

using binary exponentiation. In a general ring this operation is defined for $n \geq 0$.

The Ring of Integers $\mathbb{Z}$

The ring of integers is the structure

$$(\mathbb{Z}, +, \cdot, 0, 1).$$

It provides the usual arithmetic operations and is also the setting for basic number-theoretic algorithms such as greatest common divisor and primality testing.

Greatest Common Divisor

The greatest common divisor is computed using the Euclidean algorithm. For integers a, b , it is characterized by

$$\gcd(a, b) = \gcd(b, a \bmod b),$$

with terminal cases

$$\gcd(a, 0) = |a|, \quad \gcd(0, b) = |b|, \quad \gcd(0, 0) = 0.$$

Primality Testing

Primality testing is based on the Miller–Rabin test. For integers in the 64-bit range, a fixed set of bases makes the test deterministic. Small factors are first removed by trial division over

$$\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\},$$

and then Miller–Rabin is run with that same set of bases. For $n < 2^{64}$, this set of bases is sufficient to guarantee determinism [1, 2].

Modular Rings and Polynomial Rings

The modular ring $\mathbb{Z}/n\mathbb{Z}$ uses addition and multiplication modulo n . It is a finite ring, and when n is prime it is a field.

Given a coefficient ring R , the polynomial ring $R[X]$ consists of finite formal sums

$$a_0 + a_1X + \cdots + a_dX^d, \quad a_i \in R.$$

Addition is coefficientwise, and multiplication is given by convolution:

$$(fg)_k = \sum_{i+j=k} f_i g_j.$$

Bibliography

- [1] Gerhard Jaeschke. On strong pseudoprimes to several bases. *Mathematics of Computation*, 61(204):915–926, 1993.
- [2] Jonathan P. Sorenson and Jonathan Webster. Strong pseudoprimes to twelve prime bases. *Mathematics of Computation*, 86(307):985–1003, 2015.